

Integrating rational functions - 3.4

Recall: $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$

$$\int \frac{1}{x-a} dx = \ln|x-a| + C$$

This allows us to integrate more complicated rational functions, i.e. functions of the form $\frac{p(x)}{q(x)}$ where p and q are polynomials.

Example : $\int \frac{1}{x^2+1} - \frac{1}{x-1} dx$

$$= \arctan(x) - \ln|x-1| + C$$

$$\frac{1}{x^2+1} - \frac{1}{x-1} = \frac{x-1}{(x^2+1)(x-1)} - \frac{x^2+1}{(x^2+1)(x-1)}$$

$$= \frac{-x^2+x-2}{x^3-x^2+x-1}, \quad \text{the same but presented as a single rational function}$$

Example : Find $\int \frac{-x^2+x-2}{x^3-x^2+x-1} dx$.

Strategy : decompose $\nearrow$ somehow into smaller pieces we can integrate!